EXPERIMENT NO:08

CODE:

```
import numpy as np
import matplotlib.pyplot as plt
from sklearn.model_selection import train_test_split
from sklearn.linear_model import LinearRegression
from sklearn.metrics import mean_squared_error, r2_score

# Dataset
X = np.array([1, 2, 3, 4, 5, 6, 7, 8, 9, 10]).reshape(-1, 1)
y = np.array([10, 20, 25, 35, 40, 50, 55, 65, 70, 80])

# Split dataset
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42
)

# Create and train model
model = LinearRegression()
model.fit(X_train, y_train)

# Predict values
y_pred = model.predict(X_test)

# Display results
print("Slope:", model.coef_[0])
```

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print("Intercept:", model.intercept_)
print("Mean Squared Error:", mean_squared_error(y_test, y_pred))
print("R2 Score:", r2_score(y_test, y_pred))

# Predict for a new value
new_value = np.array([[11]])
prediction = model.predict(new_value)
print("Prediction for X = 11:", round(prediction[0], 2))

# Graph 1: Actual data and regression line
plt.figure(figsize=(8, 5))
plt.scatter(X, y, color='blue', label='Actual Data')
plt.plot(X, model.predict(X), color='red', label='Regression Line')
plt.xlabel("Hours Studied")
plt.ylabel("Marks Scored")
plt.title("Linear Regression")
plt.legend()
plt.grid(True)
plt.show()

# Graph 2: Actual vs Predicted values
plt.figure(figsize=(7, 5))
plt.scatter(y_test, y_pred, color='green')
plt.plot([min(y), max(y)], [min(y), max(y)],
        color='red', linestyle='--')
plt.xlabel("Actual Marks")
```

```
plt.ylabel("Predicted Marks")
plt.title("Actual vs Predicted Values")
plt.grid(True)
plt.show()
```

OUTPUT:

